Band-1 Hadamard Order 6628 — Miyamoto (Double Paley-I H_1656 + C_1658 over prime 1657) + Paley-II; IN SPK via Paley-II; Gap 2052 Still OPEN

H_828/H_1656 Double PASS; C_1658 PASS; H_6628 Miyamoto PASS; H_6628 Paley-II PASS (q=3313) → IN SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6484_Miyamoto`; `2026-09-15_Band1_Hadamard_6436_Miyamoto`; `2026-09-15_Band1_Hadamard_6404_Kronecker`.
Lab. `/workspace/experiments/band1/hadamard_6628_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 828, 1656, and 6628 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double(Paley-I) seed H_{1656} = Sylvester-double(H_{828} = Paley-I($q = 827$)) and symmetric Paley conference C_{1658} over prime $q = 1657 \equiv 1 \pmod 4$, **and** via classical Paley-II H_{6628} = Paley-II($q = 3313$) (prime $\equiv 1 \pmod 4$). Classical SPK paths: Paley-I for 6628 **FAIL**s ($q = 6627 = 3 \cdot 47^2$ not prime); Kronecker **FAIL**s (4×1657); Paley-II **PASS**es. Therefore order **6628 is IN SPK via Paley-II**. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under Sylvester∪Paley∪Kronecker (SPK) closure. `clay_closed = false`. Skip $q = 1637 \to n = 6548$ (no easy seed). Prior SPK seeds **1620** / **1608** / **6436** retained; Miyamoto-only bonuses **6484/6404/6388/6196/...** excluded.

Abstract

Finite constructions of orders **828**, **1656**, and **6628**.

- H_{828} = Paley-I ($q = 827$)
- H_{1656} = Sylvester-double(H_{828}) — preferred Miyamoto seed
- C_{1658} = Paley conference over prime 1657
- H_{6628} = Miyamoto(H_{1656}, C_{1658})
- H_{6628} = Paley-II ($q = 3313$) — classical SPK path
- Paley-I for 6628: **SKIP**

SPK: 6628 IN SPK via Paley-II ($q = 3313$). Seed 1656 enters SPK via Double(Paley-I). Prior SPK **1620** / **1608** / **6436** retained in ONBOX.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6628.py`

Results. overall_pass=true; units 84/84; wall 474.9744s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 6628 = 4 \cdot 1657$; 1657 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1656} and C_{1658} (Paley conference over prime 1657).
- Seed: Double(Paley-I) half=828 → Paley-I $q = 827$ (prime $\equiv 3 \pmod 4$) → H_{828}, then Sylvester-double → H_{1656}.
- Also eligible: Paley-II $q = 3313$ (prime $\equiv 1 \pmod 4$); $2(q+1) = 6628$ → **IN SPK**.
- SPK: Paley-I $q = 6627$ **FAIL**; Paley-II $q = 3313$ **PASS**; Kronecker **FAIL**.
- Skip: $q = 1637 \to n = 6548$ (no easy seed).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_828 Paley-I	Paley-I q=827	0	0.6497
H_1656 Double	Sylvester-double(H_828)	0	2.0011
C_1658	Paley conf q=1657	0	3.2863
H_6628 Miyamoto	Miyamoto(H_1656 Double, C_1658)	0	227.715
H_6628 Paley-II	Paley-II q=3313 (classical SPK)	0	234.857
H_6628 Paley-I	SKIP q=6627=$3 \cdot 47^2$ not prime	—	—

SPK closure note

Seeds include prior SPK ships through **6484** lineage (**1620** Paley-I/II; **1608** Paley-I; **6436** Paley-II; **NOT** 6484/6404/6388/6196) + verified **1656** (Double Paley-I) + verified **6628** (Paley-II); **exclude** Miyamoto-only bonuses including **6484/6404/6388/6196/5956/5812/5476**. Known count 1362. Next constructive gap **2052** (OPEN).

Quantity	Value
6628 in SPK	**True** (Paley-II q=3313)
1656 in SPK	**True** (Double Paley-I)
1620 in SPK	**True** (prior Paley-I / Paley-II)
6436 in SPK	**True** (prior Paley-II)
1608 in SPK	**True** (prior Paley-I)
6484 / 6404 / 6388 in SPK	**False** (prior Miyamoto-only)
Next constructive gap	**2052** (OPEN)

Next Miyamoto targets (after q=1657; skipped 1637→6548)

q	n=4q	need H	build notes
1669	6676	1668	[‘PaleyI(q=1667)’]
1681	6724	1680	[‘Double(PaleyI half=840)’, ‘Kronecker(4x420)’]
1693	6772	1692	[]
1697	6788	1696	[‘Kronecker(4x424)’]
1709	6836	1708	[‘PaleyII(q=853)’]
1721	6884	1720	[‘Double(PaleyI half=860)’]

Honesty

clay_closed = false · GATE-04 LOCKED · Hadamard conjecture OPEN · gap 2052 OPEN · no RH/P!=NP/YM/Frankl/NS/AC/Clay-closed claims · **6628 IN SPK via Paley-II**.